

CMP370 Internet of Things

Unit 1-Introduction to IOT

Course Information

□ Lectures:

- Monday: 12:10-1:40 PM (Day)
- Tuesday: 7:00-8:30 AM (Morning) & 8:30-10:00 AM (Day)
- Wednesday: 7:00-8:30 AM (Morning) & 12:10-1:40 PM (Day)
- Thursday: 10:40-12:10 PM (Morning)
- Friday: :8:30-10:00 AM (Day) & 10:40-12:10 AM (Morning)

□ Assignments

□ Case Studies

□ Lab. Exercises

Syllabus

Contents in Detail

Specific Objectives	Contents
Grasp the ideas of the Internet of Things (IoT), including its characteristics, trends and its applications in various sectors.	Unit 1 : Introduction to IoT (3 hrs) 1.1 Overview of IoT: Definition and Characteristics of IoT. 1.2 IoT Evolution and trends 1.3 Applications of IoT in various domains: Smart cities, healthcare, industry, and agriculture.
Understand the IoT architecture and communication protocols used in IoT systems, which is essential for IoT application development, deployment, and maintenance.	Unit 2: IoT Architecture and communication Protocols (6 hrs) 2.1 IoT architecture (Device layer, Network layer, Cloud layer). 2.2 IoT reference model (4-layer and 7-layer models) 2.3 Communication models (Request-response, Publish-subscribe) 2.4 IoT protocols (MQTT, CoAP, HTTP, WebSocket)
Understand about the sensor and actuators and its types along with its characteristics	Unit 3: Sensors, Actuators, and Embedded Systems (5 hours) 3.1 Types of sensors and its characteristics (temperature, humidity, motion etc.) 3.2 Actuators (relays, motors) 3.3 Types of Microcontrollers and embedded systems in IoT (like Raspberry Pi, Arduino)

Syllabus

• Able to design and develop IoT projects using Embedded Systems and programming languages such as Python with the help of IDEs and debugging tools. • Gain practical experience with Raspberry Pi by setting up the environment, working with GPIOs and sensors, and building IoT System.	Unit 4: Developing IoT System: Interfacing to connect with external devices using development boards (10 hrs) 4.1 Role of microcontrollers (Raspberry Pi) 4.2 Introduction to Python for IoT (basics of Python and libraries like RPi.GPIO, smbus, spidev). 4.3 Setting up the Raspberry Pi for IoT 4.4 Sensor Interfacing with Python 4.5 Actuator Interfacing with Python 4.6 Data Logging and Visualization 4.7 IoT Communication (MQTT) 4.8 Case study: Home automation, Weather Monitoring System.
• Understand the Cloud and its relationship with (IoT). It deals with the importance of Fog computing for the Internet of Things.	Unit 5: Cloud Computing in IoT (3 HRS) 5.1 Role of Cloud Computing in IoT. 5.2 Study of IoT Cloud platforms(ThinkSpeak, Google Cloud) 5.3 Emerging technologies.
• Understand the security challenges and opportunities and best practices in IoT System	Unit 6: Securing in IoT System (3 HRS) 6.1 IoT Security Challenges and Opportunities 6.2 IoT Security Practices

Practical works

SN	Implementation Description
1	Familiar with Raspberry pI boards and their I/O pins.
2	Familiar with Raspberry pI platform and IDE for programming , Debugging, and Uploading
3	Building a basic IoT project using Raspberry Pi/Arduino with sensor integration.
4	Implement communication between devices using MQTT and HTTP protocols.
5	Send sensor data to ThingSpeak or AWS IoT and visualize data using dashboards.
5	Implement encryption and secure communication in IoT systems using SSL/TLS.
6	Perform basic data analysis on IoT data and visualize results using a Python library (e.g., Matplotlib).
7	Implement an edge computing system for real-time data processing on a Raspberry Pi.
8	Develop an end-to-end IoT application (e.g., Smart home automation, Weather monitoring system - Temp, Humidity, Smart parking system, Smart irrigation system, Smart lock based on face recognition)

Practical and References

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		30	Semester-End examination	50
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		20		
Attendance & Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			
Viva	30%			
Total Internal		50		
Full Marks: 50 + 50 = 100				



8. Prescribed Books and References

Text Books

References

1. IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things, David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome Henry, Cisco Press, 2017
2. Introduction to IoT:-Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Cambridge University Press
3. Digital Electronics with Arduino: Learn How To Work With Digital Electronics And Microcontrollers by Bob Dukish, BPB Publications (2020)
4. Programming Arduino: Getting Started with Sketches (Tab) 2nd Edition by Simon Monk, McGraw Hill TAB (2016)
5. The Official Raspberry Pi Beginner's Guide 4th Edition (The Official Raspberry Pi Beginner's Guide: How to use your new computer) by Gareth Halfacree, Raspberry Pi Press (2020)
6. Programming the Raspberry Pi, Third Edition: Getting Started with Python 3rd Edition by Simon Monk, McGraw Hill TAB (2021)

Course Objectives

- To introduce students to basic concepts of Internet of Things (IoT).
- To familiarize students with various IoT platforms and devices.
- To develop students' understanding of data acquisition, processing, and visualization techniques.
- To enable students to choose appropriate devices and platforms for specific IoT applications.
- To provide practical knowledge in interfacing sensors and actuators for real-time data acquisition.
- To Encourage students to apply IoT technology to solve real-world problems.

Internet

- “Internet” refers to the vast category of applications and protocols built on top of sophisticated and interconnected computer networks, serving billions of users around the world in 24/7 fashion.
- It enables people to access websites, email, stream media, and use various online services.
- The Internet relies on a complex infrastructure of cables, servers, routers, and other technology to facilitate the transmission of data.



Evolutionary Phase of Internet

Internet Phase	Definition
Connectivity (Digitize access)	This phase connected people to email, web services, and search so that information is easily accessed.
Networked Economy (Digitize business)	This phase enabled e-commerce and supply chain enhancements along with collaborative engagement to drive increased efficiency in business processes.
Immersive Experiences (Digitize interactions)	This phase extended the Internet experience to encompass widespread video and social media while always being connected through mobility. More and more applications are moved into the cloud.
Internet of Things (Digitize the world)	This phase is adding connectivity to objects and machines in the world around us to enable new services and experiences. It is connecting the unconnected.

Internet of Things (IoT)-Introduction

- **Internet of Things (IoT)** comprises things that have unique identities and are connected to the Internet.
- IoT refers to the network of interconnected devices (sensors, actuators, appliances) that communicate and exchange data over the internet to perform tasks autonomously or with minimal human intervention.
- While many existing devices, such as networked computers or 4G-enabled mobile phones, already have some form of unique identities and are also connected to the Internet, the focus on IoT is in the configuration, control and networking via the Internet of devices or "things" that are traditionally not associated with the Internet.
- These include devices such as thermostats, utility meters, a bluetooth-connected headset, irrigation pumps and sensors, or control circuits for an electric car's engine.

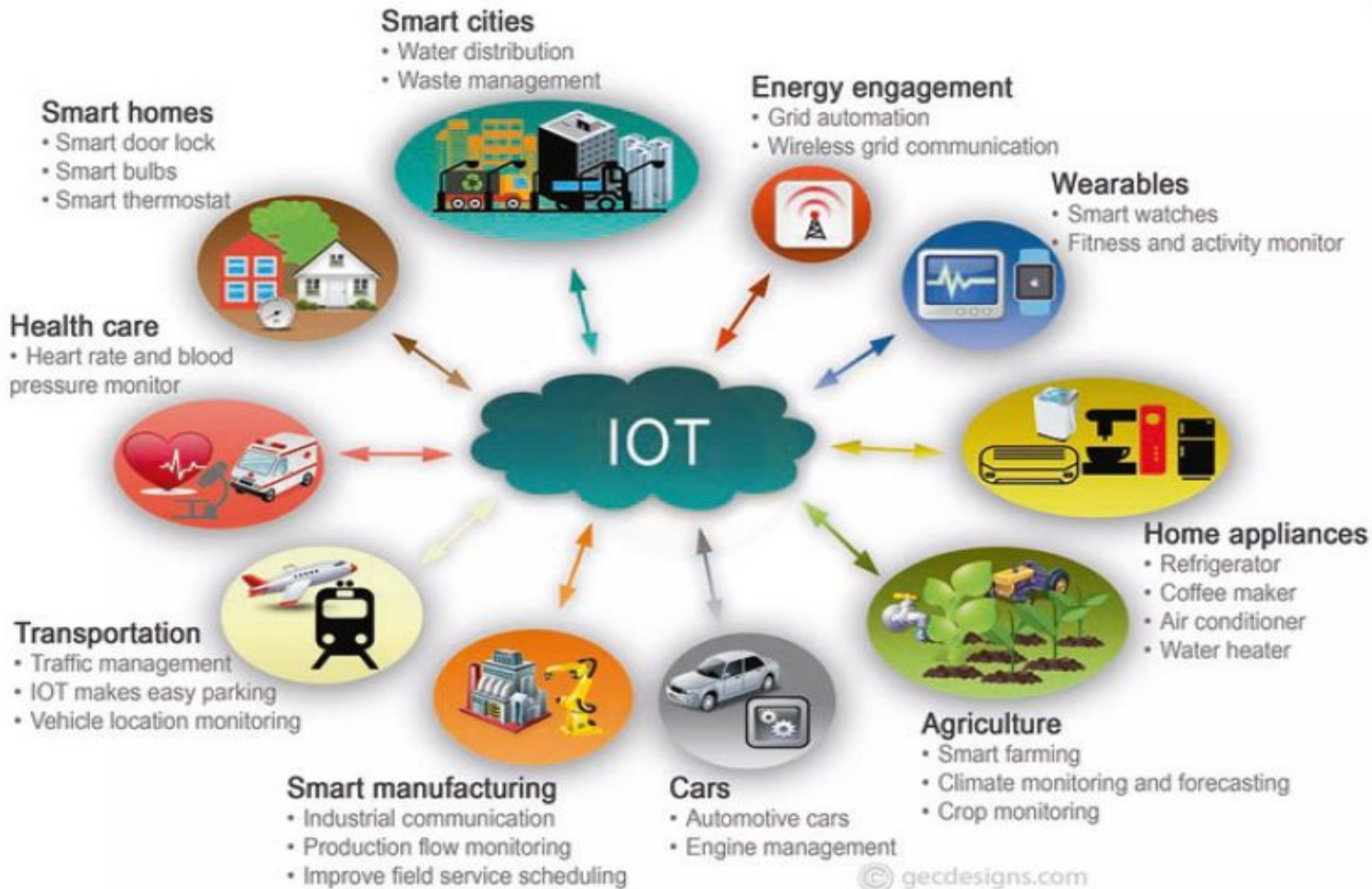
IOT-Definition

A dynamic global network infrastructure with self-configuring capabilities based on standard and interoperable communication protocols where physical and virtual "things" have identities, physical attributes, and virtual personalities and use intelligent interfaces, and are seamlessly integrated into the information network, often communicate data associated with users and their environments.

Source: Bahga & Madisetti, © 2014

ITU, ISO, and Cisco defined the Internet of Things as follows:

“A global infrastructure for the information society, enabling advanced services by interconnecting (physical and virtual) things based on existing and evolving interoperable information and communication technologies.”



Internet of Everything

- The definition of IOT focuses mostly on connectivity and sensory requirements for entities involved in typical IoT environments.
- Whereas those definitions reflect IoT's basic requirements, new IoT definitions give more value to the need for ubiquitous and autonomous networks of objects where identification and service integration have an important and inevitable role.
- Internet of Everything (IoE) is used by Cisco to refer to people, things, and places that can share their services to other entities.
- The main difference between IoT and IoE is that IoT only concerns itself with the non-human aspects of technology, while IoE consists of all the other factors, which include machine-to-people (M2P) and technology-assisted peer-to-peer (P2P) interactions in addition to the features of IoT.

IOT Prospects

- Internet of Things is a new revolution in the capabilities of the endpoints that are connected to the Internet, and is being driven by the advancements in capabilities (in combination with lower costs) in sensor networks, mobile devices, wireless communications, networking and cloud technologies.
- IoT is not a single technology, it's a combination of technologies and domain knowledge.
- As a result, engineers from different domains have to work together for building a complete IoT product.
- The latest figures indicate that there were 18.8 billion connected IoT devices globally by the end of 2024, reflecting a 13% growth from 16.6 billion in 2023
- IoT Analytics projects 40 billion connected IoT devices by 2030.

Scope of IOT

- Connecting things (devices, appliances, machines) to the Internet.
- IoT allows these things to communicate and exchange data (control & information, that could include data associated with users) while executing meaningful applications towards a common user or machine goal.
- Data itself does not have a meaning until it is contextualized processed into useful information.
- Applications on IoT networks extract and create information from lower level data by filtering, processing, categorizing, condensing and contextualizing the data.
- This information obtained is then organized and structured to infer knowledge about the system and/or its users, its environment, and its operations and progress towards its objectives, allowing a smarter performance.

Inferring information and knowledge from data

Data

- Raw and unprocessed data obtained from IoT devices/systems.

Information

- Information is inferred from data by filtering, processing, categorizing, condensing and contextualizing data.

Knowledge

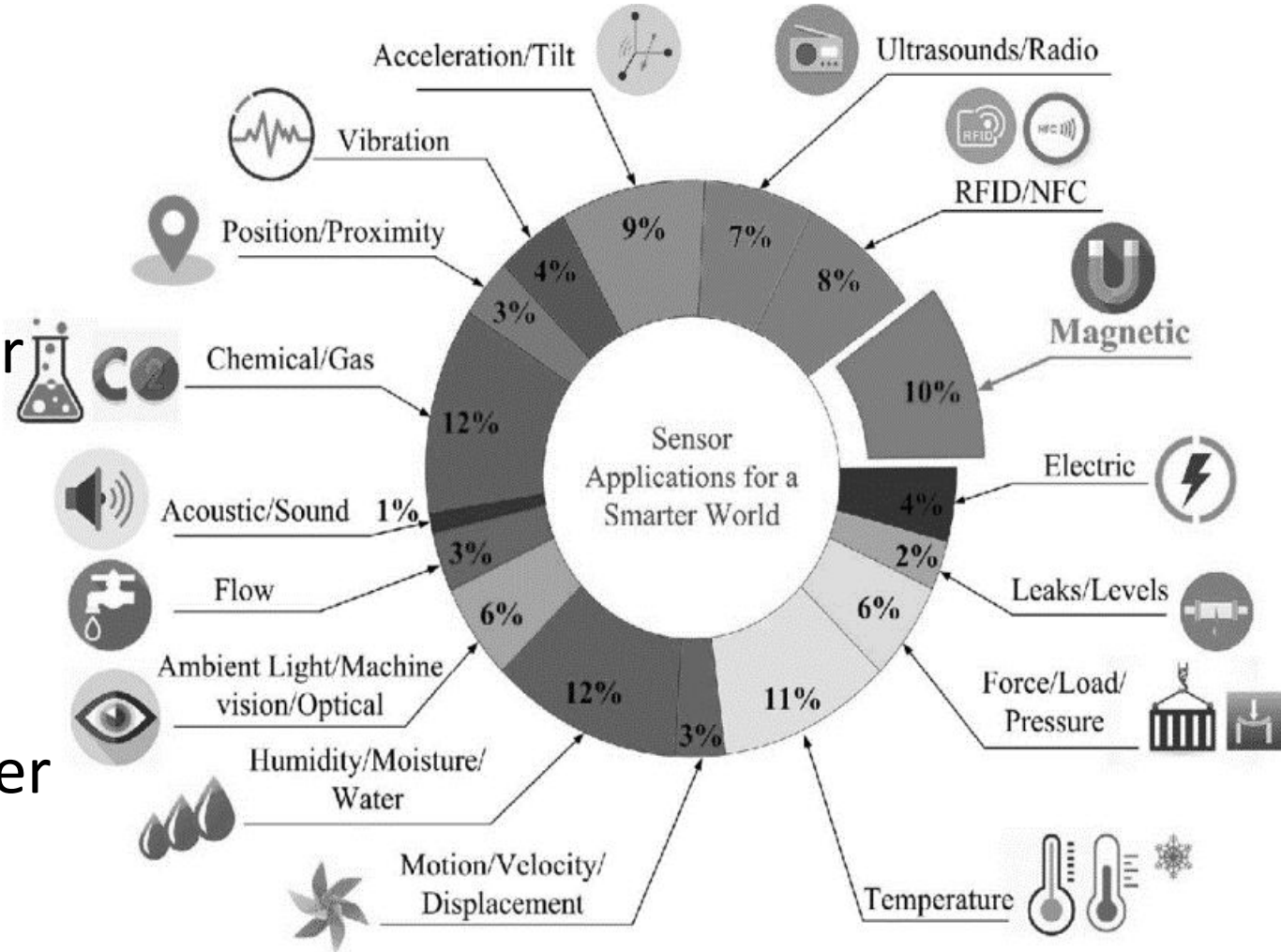
- Knowledge is inferred from information by organizing and structuring information and is put into action to achieve specific objectives.

“Things” in IoT

- In the IoT, things refer to a variety of devices which have unique identities and can perform remote sensing, actuating and monitoring capabilities.. It can be anything even humans in it become a thing.
- For something to qualify as a “thing”, it requires identity of its existence.
- The “thing” in a network can be monitored/measure. For example, a temperature sensor could be a thing.
- Things are capable of exchanging data with other connected devices in the system.
- The data could be stored in a centralized server (or cloud), processed there and a control action could be initiated.
- The devices having all the above characteristics are known as things.

“Things” in IoT

- Some of the famous “things” are temperature sensors, pressure sensors, humidity sensors, etc.
- The data from these sensors are collected and sent it to the cloud or stored it in local server for data analysis.
- Based on the data analysis, the control action would be taken.
- For example, switching off the water heater remotely when the water is heated as per requirement.



“Things” in IoT

Not just sensors, the following can also be called as things:

- Industrial motors
- Wearables (e.g., watch)
- Vehicles
- Shoes
- Heart monitoring implants (e.g., pacemaker, ECG real-time tracking)
- Biochip transponders (for animals in farms)
- Automobiles with built-in sensors (automobile feature real-time monitoring)
- Food/perishables quality measuring



Characteristics of IoT

- 1. Dynamic & Self-Adapting**
- 2. Self-Configuring**
- 3. Interoperable Communication Protocols**
- 4. Unique Identity**
- 5. Integrated into Information Network**
- 6. Intelligence**
- 7. Scalability**
- 8. Heterogeneity**

1. Dynamic & Self-Adapting

- IoT devices and systems may have the capability to dynamically adapt with the changing contexts and take actions based on their operating conditions, user's context, or sensed environment.
- For example, consider a surveillance system comprising of a number of surveillance cameras. The surveillance cameras can adapt their modes (to normal or infra-red night modes) based on whether it is day or night. Cameras could switch from lower resolution to higher resolution modes when any motion is detected and alert nearby cameras to do the same.
- In this example, the surveillance system is adapting itself based on the context and changing (e.g., dynamic) conditions.

2. Self-Configuring:

- IoT devices may have self-configuring capability, allowing a large number of devices to work together to provide certain functionality (such as weather monitoring).
- These devices have the ability configure themselves (in association with the IoT infrastructure), setup the networking, and fetch latest software upgrades with minimal manual or user intervention.

3. Interoperability

- IoT devices may support a number of interoperable communication protocols and can communicate with other devices and also with the infrastructure.
- IoT devices and systems are designed to work together seamlessly, even if they are developed by different manufacturers or operate on different platforms.
- Interoperability allows for greater flexibility and scalability in building IoT solutions.

4. Unique Identity:

- Each IoT device has a unique identity and a unique identifier (such as an IP address or a Uniform Resource Identifier (URI)).
- IoT device interfaces allow users to query the devices, monitor their status, and control them remotely, in association with the control, configuration and management infrastructure.
- Each IOT device has an IP address. This identity is helpful in tracking the equipment and at times to query its status.

5. Integrated into Information Network

- IoT devices are usually integrated into the information network that allows them to communicate and exchange data with other devices and systems.
- IoT devices can be dynamically discovered in the network, by other devices and/or the network, and have the capability to describe themselves (and their characteristics) to other devices or user applications.

6. Intelligence

- Data collection is not enough in IoT, extraction of knowledge from the generated data is very important.
- For example, sensors generate data, but that data will only be useful if it is interpreted properly.
- Intelligence is one of the key characteristics in IoT.

7. Scalability

- The number of elements (devices) connected to IoT zone is increasing day by day. Therefore, an IoT setup should be capable of handling the expansion.
- It can be either expand capability in terms of processing power, Storage, etc. as vertical scaling or horizontal scaling by multiplying with easy cloning

8. Heterogeneity

- The devices in the IoT are heterogeneous as based on different hardware platforms and networks.
- They can interact with other devices different networks.

Predecessors of IoT

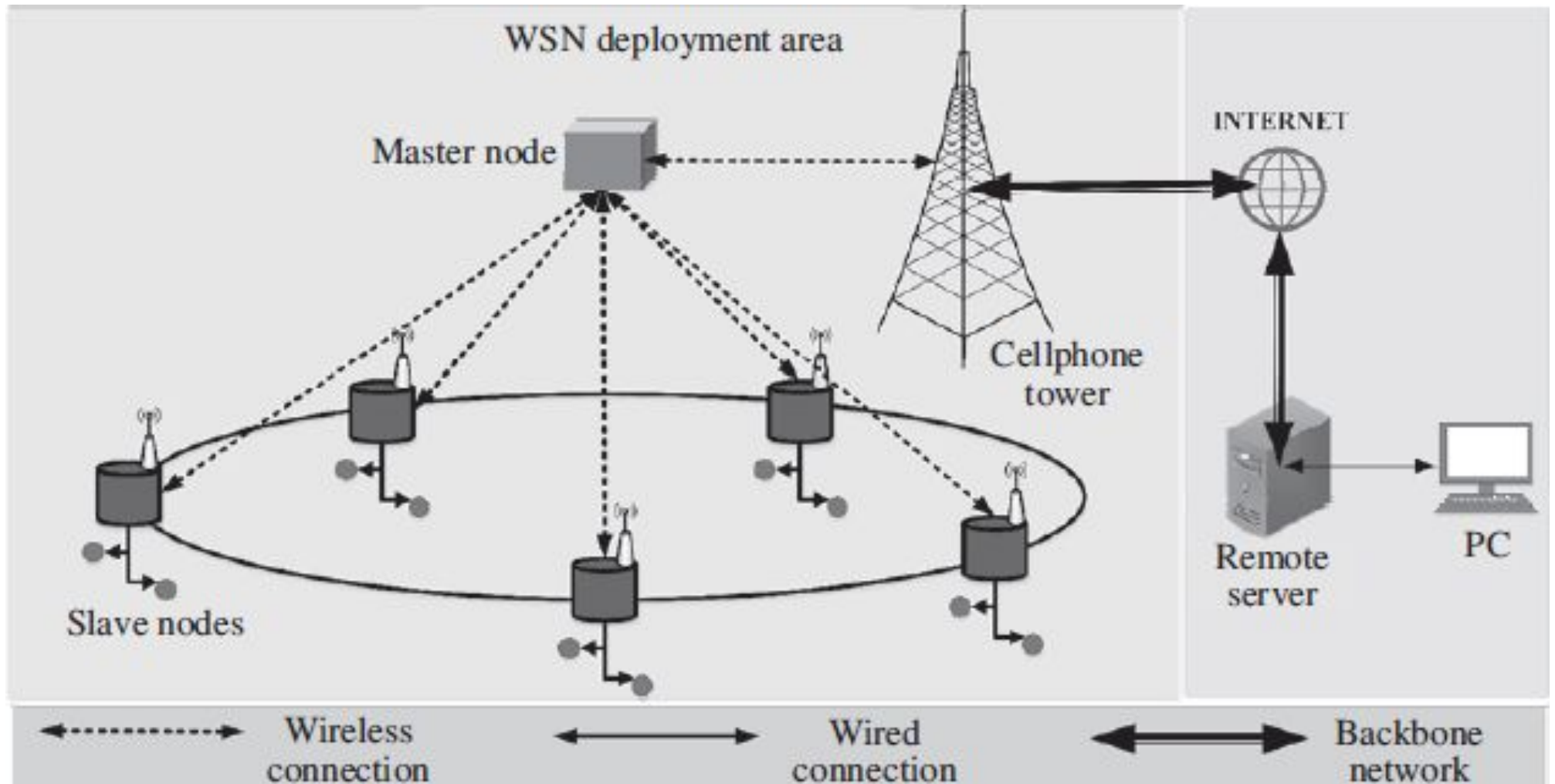
- Wireless sensor networks (WSN)
- Machine-to-machine (M2M) communications
- Cyber physical systems (CPS).

Initially, Radio-Frequency Identification (RFID) used to be the dominant technology behind IoT development, but with further technological achievements, wireless sensor networks (WSN) and Bluetooth-enabled devices augmented the mainstream adoption of the IoT trend.

Wireless sensor networks (WSN)

- WSN is a networking paradigm that makes use of spatially distributed sensors for gathering information concerning the immediate environment of the sensors and collecting the information centrally.
- The sensors are not standalone devices but a combination of sensors, processors, and radio units—referred to as sensor nodes—sensing the environment and communicating the sensed data wirelessly to a remote location, which may or may not be connected to a backbone network.

WSN Deployment



Machine-to-Machine Communications

- The machine-to-machine (M2M) paradigm, as the name suggests, implies a system of communication between two or more machines/devices without human intervention.
- Some basic examples of M2M communication in our daily lives include ATM machines signaling banks about the need to refill them with cash.
- 3GPP which is responsible for unifying and benchmarking telecommunication standards across seven different telecommunication standardization organizations, refers to M2M as machine type communications (MTC).

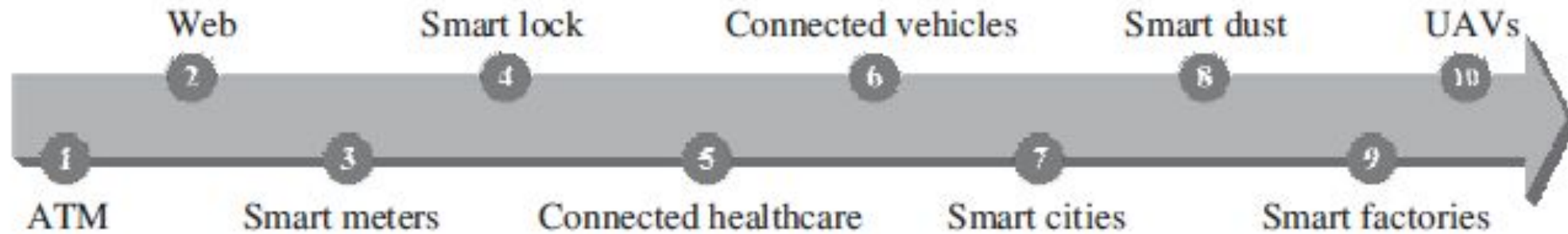
Cyber Physical Systems

- Cyber physical systems (CPS) are Internet-based and networked monitoring and controlling systems that are regulated and governed by feedback-based intelligent control algorithms.
- They are mainly designed to monitor and control physical world processes linked to businesses and industries
- The most interesting aspect of CPS is the involvement of the concept of human-in-the-loop, which is an integral part of many CPS-based solutions.
- The striking difference of CPS from paradigms such as WSN and M2M is the inclusion of a compulsory feedback system.

Emergence of IoT

- The modern-day advent of network-connected devices has given rise to the popular paradigm of the Internet of Things (IoT).
- Kevin Ashton is accredited for using the term “Internet of Things” for the first time during a presentation in 1999 on supply-chain management. At the time, he was an executive director at MIT’s Auto-ID Center, where he contributed to the extension of RFID applications into broader domains, which built the foundation for the current IoT vision.
- IoT is an anytime, anywhere, and anything network of Internet-connected physical devices or systems capable of sensing an environment and affecting the sensed environment intelligently.
- This is generally achieved using low-power and low-form-factor embedded processors on-board the “things” connected to the Internet.
- These things are made up of sensors/actuators and processors, which connect to the Internet through both wired and wireless technologies.

Evolution of IoT



- **ATM:** ATMs or automated teller machines are cash distribution machines, which are linked to a user's bank account. ATMs dispense cash upon verification of the identity of a user and their account through a specially coded card.
- **Web:** World Wide Web is a global information sharing and communication platform. The Web became operational for the first time in 1991. Since then, it has been massively responsible for the many revolutions in the field of computing and communication.

Evolution of IoT

- **Smart Meters:** The earliest smart meter was a power meter, which became operational in early 2000. These power meters were capable of communicating remotely with the power grid. They enabled remote monitoring of subscribers' power usage and eased the process of billing and power allocation from grids.
- **Digital Locks:** Digital locks can be considered as one of the earlier attempts at connected home-automation systems. Present-day digital locks are so robust that smartphones can be used to control them.
- **Connected Healthcare:** Here, healthcare devices connect to hospitals, doctors, and relatives to alert them of medical emergencies and take preventive measures. The devices may be simple wearable appliances, monitoring just the heart rate and pulse of the wearer, as well as regular medical devices and monitors in hospitals.

Evolution of IoT

- **Connected Vehicles:** Connected vehicles may communicate to the Internet or with other vehicles, or even with sensors and actuators contained within it. These vehicles self-diagnose themselves and alert owners about system failures.
- **Smart Cities:** This is a city-wide implementation of smart sensing, monitoring, and actuation systems. The city-wide infrastructure communicating amongst themselves enables unified and synchronized operations and information dissemination.
- **Smart Dust:** These are microscopic computers. Smaller than a grain of sand each, they can be used in numerous beneficial ways, where regular computers cannot operate. For example, smart dust can be sprayed to measure chemicals in the soil or even to diagnose problems in the human body.

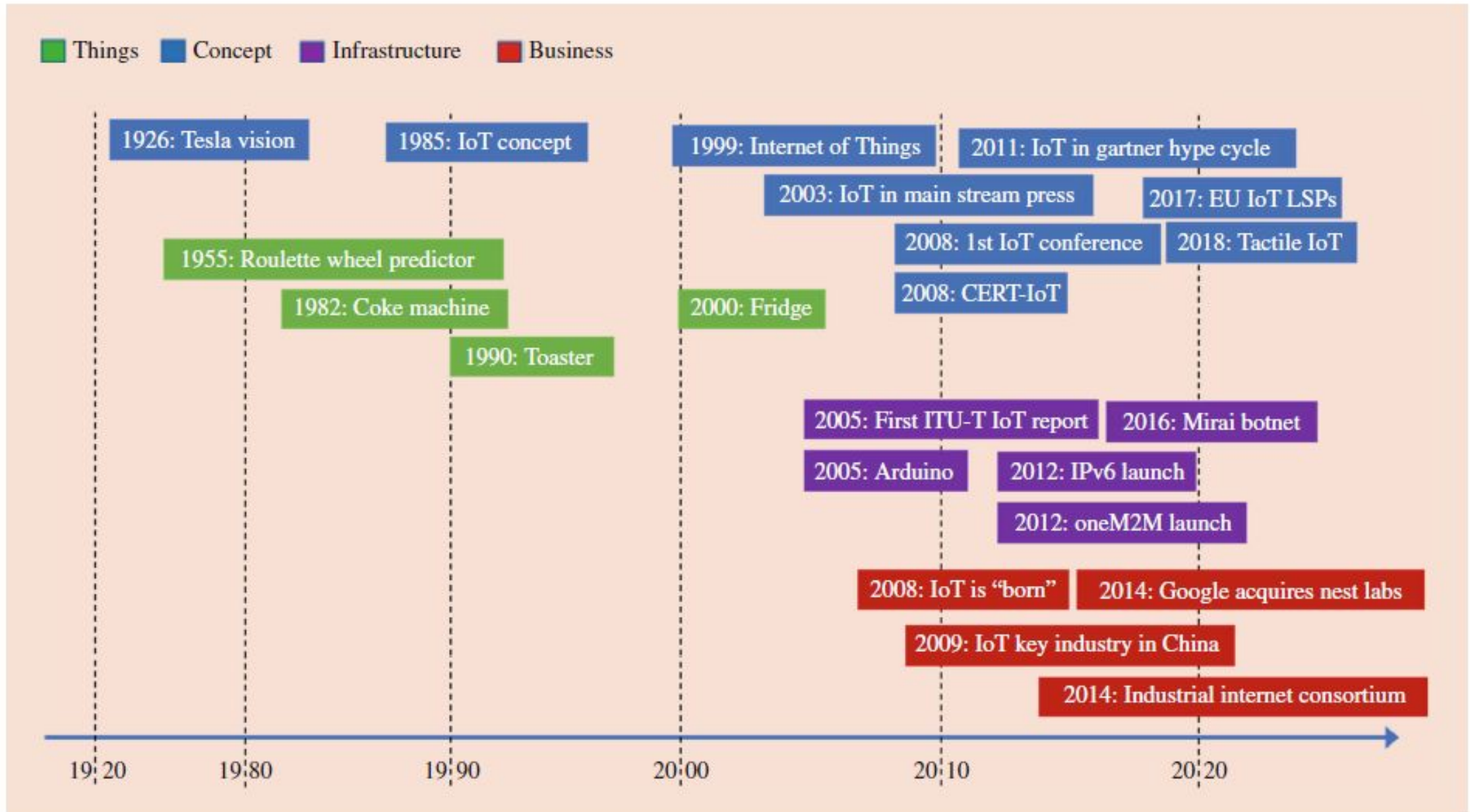
Evolution of IoT

- **Smart Factories:** These factories can monitor plant processes, assembly lines, distribution lines, and manage factory floors all on their own. The reduction in mishaps due to human errors in judgment or unoptimized processes is drastically reduced.
- **UAVs:** UAVs or unmanned aerial vehicles have emerged as robust public domain solutions tasked with applications ranging from agriculture, surveys, surveillance, deliveries, stock maintenance, asset management, and other tasks.

IoT versus M2M

- Telecommunication services providers introduced the term M2M, and technically emphasized on machine interactions via one or more communication networks (e.g., 3G, 4G, 5G, satellite, public networks).
- M2M is part of the IoT and is considered as one of its sub-domains.
- In terms of operational and functional scope, IoT is vaster than M2M and comprises a broader range of interactions such as the interactions between devices/things, things, and people, things and applications, and people with applications.

Timeline of IoT history (Optional)



IoT versus Cyber Physical System (CPS)

- Cyber physical systems (CPS) encompasses sensing, control, actuation, and feedback as a complete package.
- In other words, a digital twin is attached to a CPS-based system.
- Digital twin is a virtual system–model relation, in which the system signifies a physical system or equipment or a piece of machinery, while the model represents the mathematical model or representation of the physical system's behavior or operation.
- Based on feedback from the digital twin, a physical system can be easily given corrective directions/commands to obtain desirable outputs. In contrast, the IoT paradigm does not compulsorily need feedback or a digital twin system. IoT is more focused on networking than controls.

IOT Trends

- **Big data**
- **Cloud/fog/edge computing**
- **5G and beyond**
- **Artificial intelligence (AI)/Machine learning (ML)**
- **Cognitive communication networks**
- **Network function virtualization (NFV)**
- **Software Defined Network (SDN)**

Big Data

- The network traffic or data is classified as big data if it satisfies specific characteristics of 1) volume, 2) variety, 3) value, 4) velocity, and 5) veracity.
- Big data analytics has the potential to process data from IoT devices in real time and store them using various storage schemes.
- Once acquired, this voluminous data can be used for studying patterns in network behavior, usage, customer experiences, mobility, connectivity issues, among many other interesting features.

Cloud/fog/edge computing

- Although a significant number of IoT devices are smaller in size and resource-constrained, their massive-scale use in various applications eventually leads to a formidable amount of information to be processed and handled.
- Due to such limitations, these IoT devices need to depend on external platforms, particularly cloud/fog/edge computing schemes, to address their processing issues and generate meaningful information from the gathered data.
- The choice of platform is application dependent, that is, while cloud computing has unlimited resources, fog/edge computing reduces operational latencies significantly.
- Starting as a research paradigm, these domains have gathered worldwide acceptance and are mostly included in mainstream IoT architectures and applications.

5G and beyond

- The new 5G technologies provide services with much higher speeds, as they focus on providing ubiquitous high-speed connectivity to all device types.
- Researchers envision 5G as a driving force for IoT. The features of 5G have also started the race for beyond 5G technologies and paradigms.
- Envisioned as operating in the Terahertz band of the frequency spectrum, beyond 5G technologies are being speculated to have data rates up to Gbps.

Artificial intelligence (AI)/Machine learning (ML)

- With the deployment of numerous IoT devices and applications, the complexity and size of data over and beyond the networks have increased significantly.
- Modern-day AI/ML based tools have significantly powerful inferencing procedures, which can outperform almost any of the standard statistical methods.
- AI/ML has been used for extracting information from raw data, be it from agricultural sensors, smart home sensors, or network security analyzers.
- AI/ML typically needs a lot of data to learn and make predictions, the field is evolving, and new techniques are being created that can function with smaller datasets.

Cognitive communication networks

- Cognitive communication networks or simply, cognitive networks are capable of sensing the present network's parameters, conditions, and plans.
- Based on the sensed information, the cognition engine can devise pathways and strategies for best achieving the end goals for a certain task.

Network function virtualization (NFV)

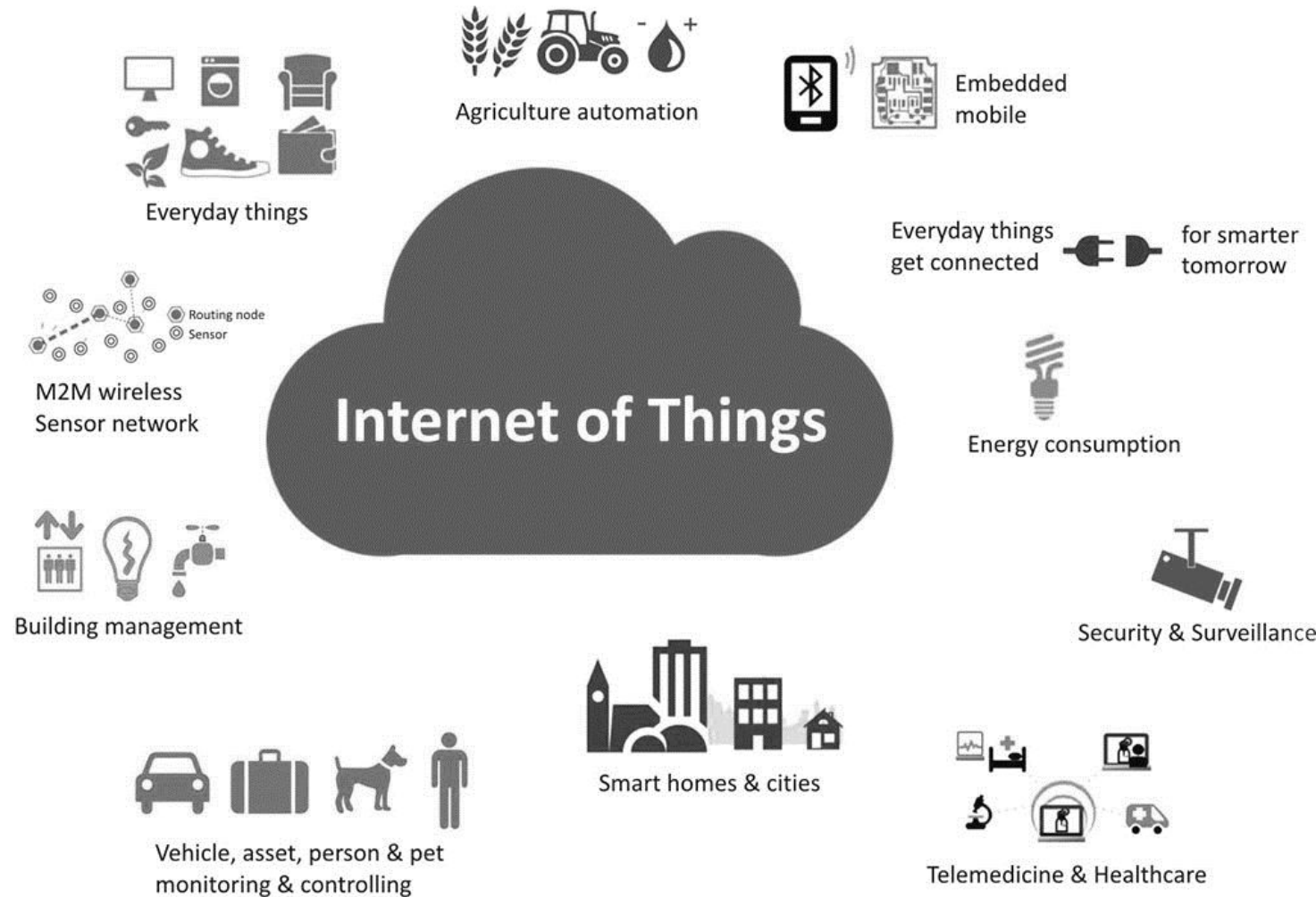
- NFV is an interesting and practical concept, which proposes the virtualization of major network elements such that software virtualizes network hardware by providing the same functionalities and added features.
- The concept of NFV arose due to the difficulty of reconfiguring (both changing and upgrading) installed network infrastructures. Physically going to every network element for changes (updates, software patches) can be a significant challenge in terms of time, money, and human resources, especially for enterprise-grade networks consisting of tens of thousands of network elements.
- NFV utilizes the concept of virtualization to provide services similar to network elements through standard servers.

Software-defined networks (SDN)

- For the seamless transition of data routes, self-organization, configuration optimization, and smart transmissions, SDN has emerged as a popular choice.
- SDN reduces the complexity of traditional networks by introducing a centralized control structure through the separation of control and data planes of network elements.
- A centralized view of the whole network ushers in the benefits of better controllability, network stability, and increased efficiencies.
- SDN introduces an entirely new way of handling network traffic by separating the control and data planes to provide a centralized architecture for the whole network.

Applications of IOT

- ❑ Agriculture
- ❑ Assets Tracking
- ❑ Energy Sector
- ❑ Defense
- ❑ Embedded Applications
- ❑ Education
- ❑ Waste Management
- ❑ Healthcare Products
- ❑ Telemedicine
- ❑ Safety And Security Sector
- ❑ Smart City Applications etc.



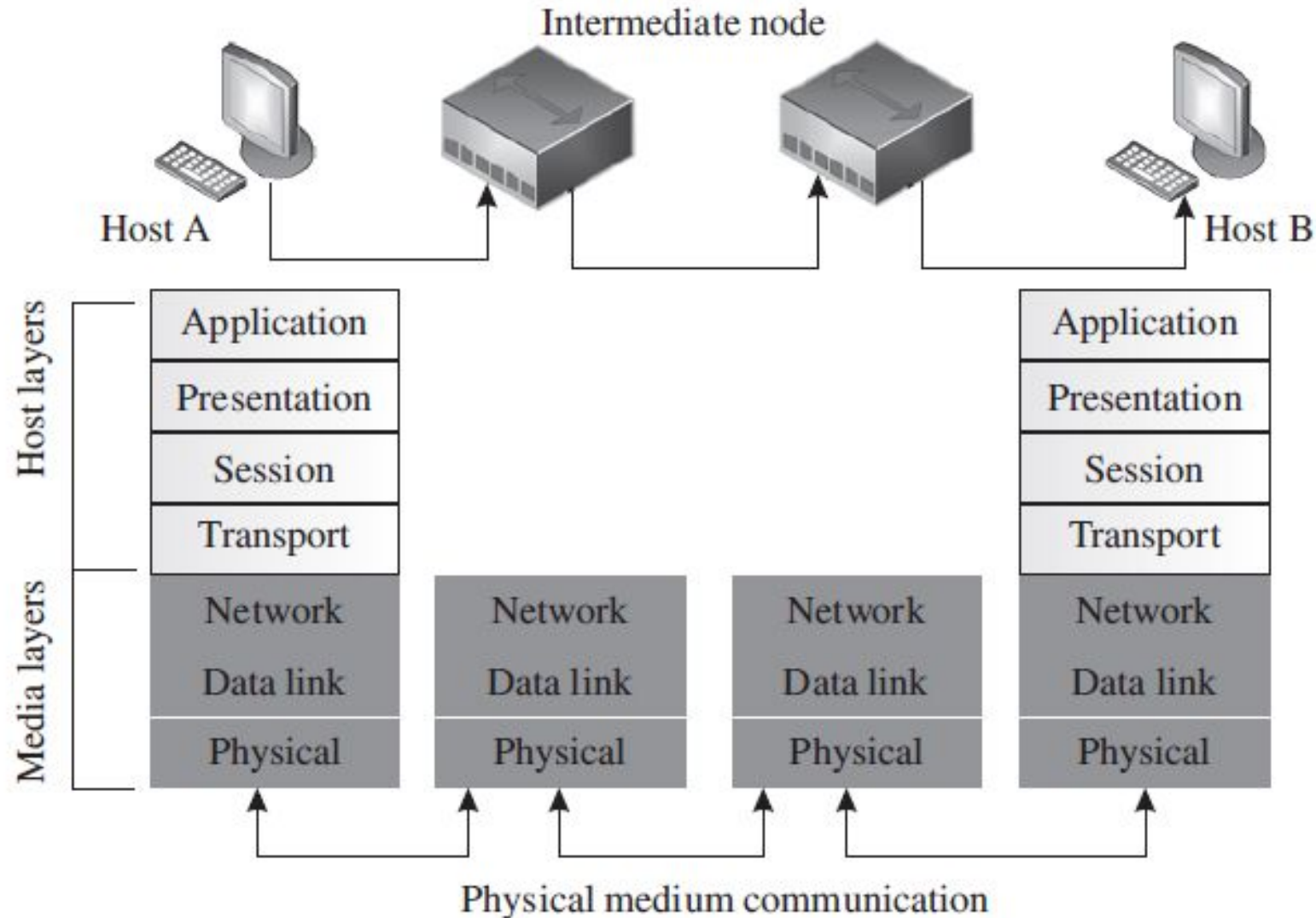
Applications of IoT

- **Smart Cities:** Traffic management (smart traffic lights), waste management (smart bins), and energy optimization (smart grids).
- **Healthcare:** Remote patient monitoring (wearable devices), smart hospitals (inventory tracking), and telemedicine.
- **Industry:** Industrial IoT (IIoT) for predictive maintenance, supply chain optimization, and factory automation.
- **Agriculture:** Precision farming (soil moisture sensors), livestock monitoring, and automated irrigation systems.
- **Other Domains:** Smart homes (lighting, security), transportation (fleet tracking), and environmental monitoring (air quality sensors).

Advantages of IoT

- **Efficient resource utilization:** IoT devices can monitor resource usage in real-time, providing data to optimize consumption and reduce waste. By automating control based on this data, IoT enables more efficient allocation and management of resources
- **Minimize human effort:** As devices of IoT interact and communicate with each other and do a lot of tasks for us, then they minimize human effort.
- **Save time:** As it reduces human effort, then it definitely saves out time. Time is the primary factor which can save through IoT platform.
- **Enhance data collection:** IoT uses connected sensors to automatically gather data from many different places and things. This allows us to collect more data, more often, and more accurately than we could before.
- **Improve security:** If we have a system where all these things are interconnected, then we can make the system more secure and efficient.

Networked communication between two hosts following the OSI model



Layer	Name	Location	PDU	Function	Examples
1	Physical	Media	Symbol	Communication over physical medium	Ethernet, FDDI, B8ZS, V.35, V.24, RJ45
2	Data link	Media	Frame	Reliability of communication over physical medium	IEEE 802.5/ 802.2, IEEE 802.3/802.2, PPP, HDLC, Frame Relay, ATM, FDDI
3	Network	Media	Packet	Structuring of data and routing between multiple nodes	DDP, IP, AppleTalk, IPX
4	Transport	Host	Segment	Reliability of communication over networks or between hosts	SPX, TCP, UDP
5	Session	Host	Data	Establishment, management, and termination of remote sessions	NetBios names, NFS, RPC, SQL
6	Presentation	Host	Data	Syntactic conversion of data and encryption	Encryption, ASCII, MIDI, PICT, JPEG, EBCDIC, TIFF, GIF, MPEG
7	Application	Host	Data	User identification, authentication, privacy, and quality of service	SNMP, Telnet, WWW browsers, HTTP, NFS, FTP

IoT Challenges

- **Scalability**
- **Security**
- **Privacy**
- **Big data and data analytics**
- **Interoperability**
- **Bandwidth**
- **Regulation**

Scalability

- The massive scale of connected devices in the Internet of Things (IoT) ecosystem presents significant challenges that impact the design, deployment, management, and security of IoT systems.
- This can lead to congestion, especially in dense environments like smart cities
- For example, In a smart city with millions of sensors (traffic lights, cameras, air quality monitors), simultaneous data transmission can overwhelm traditional networks.
- The system that stores, analyses the data from these IoT devices needs to be scalable.

Security

- With more “things” becoming connected with other “things” and people, security is an increasingly complex issue for IoT.
- If a device gets hacked, its connectivity is a major concern.
- A compromised device can serve as a launching point to attack other devices and systems. IoT security is also pervasive across just about every facet of IoT.

Privacy

- As sensors become more prolific in our everyday lives, much of the data they gather will be specific to individuals and their activities.
- This data can range from health information to shopping patterns and transactions at a retail establishment.
- For businesses, this data has monetary value.
- Organizations are now discussing who owns this data and how individuals can control whether it is shared and with whom.

Big data and data analytics

- IoT systems with lots of sensors will produce so much data that it will be difficult to handle.
- This data will provide critical information and insights if it can be processed in an efficient manner.
- The challenge, however, is evaluating massive amounts of data arriving from different sources in various forms and doing so in a timely manner.

Interoperability

- IoT is a developing technology, so there are many different protocols and designs trying to gain popularity and become the norm.
- Some of these are proprietary, and some are open.
- Although recent standards are making things better, IoT networks often have to deal with multiple protocols and implementations.

Power

- IoT devices are usually resource-constrained concerning power and computational capability.
- These devices need to last for a long time irrespective of their limited battery power.
- Need for new hardware designs that consume minimum power for operation.
- Various upcoming research solutions focus on developing high-density batteries/cells for enabling long-term use of IoT systems.
- Research in this domain includes the design of low-power processors and hardware, design of low-power consuming computation techniques and algorithms, energy harvesting, alternative sources of energy, and others.

Bandwidth

- Connectivity is a bigger challenge to the IoT.
- As the size of the IoT market grows exponentially, some experts are concerned that bandwidth-intensive IoT applications such as video streaming will soon struggle for space on the IoT's current server-client model.

Regulation

- The lack of strong IoT regulations is a big part of why the IoT remains a severe security risk, and the problem is likely to get worse as the potential attack surface expands to include ever more crucial devices.
- When medical devices, cars and children's toys are all connected to the Internet, it's not hard to imagine many potential disaster scenarios unfolding in the absence of sufficient regulation.

2G, 3G & 4G Network Architecture

